Plasma Chemistry and Plasma Processing (2024) 44:2179–2198
https://doi.org/10.1007/s11090-024-10509-0

ORIGINAL PAPER

Assessing the Preservation Effectiveness: A Comparative Study of Plasma Activated Water with Various Preservatives on *Capsicum annuum* L. (Jalapeño and Pusa Jwala)

Vikas Rathore[1] · Piyush Sharma[2] · Arun Prasath Venugopal[2] · Sudhir Kumar Nema[1,3]

Received: 22 May 2024 / Accepted: 31 August 2024 / Published online: 14 September 2024
© The Author(s) 2024

Abstract
The study investigates the efficacy of plasma-activated water (PAW) in preserving green chillies (jalapeño and pusa jwala) and compared it with various household fruits and vegetables cleaners' solutions. PAW was prepared using a pencil plasma jet with air as the plasma forming gas. The results of visual analysis revealed that PAW-treated chillies maintain their fresh appearance even after 21 days, exhibiting significantly lower spoilage compared to control (ultrapure milli-Q water) and fruits and vegetables cleaners' solutions. PAW demonstrated antimicrobial properties, effectively reducing microbial growth and spoilage on chillies over the storage period. Physical attributes, such as weight loss and firmness, are evaluated. It has been observed that PAW-treated chillies exhibit lower weight loss and higher firmness, indicating better membrane integrity and moisture retention. Microbial resistance was notably higher in PAW-treated chillies compared to control and when cleaning solutions were used. CIELAB color analysis revealed that PAW-treated chillies retain greenness, and color, freshness, outperforming control and cleaners. Sensory evaluation, including visual inspection, smell, taste, and touch, consistently favored PAW-treated chillies, emphasizing their superiority in terms of enhancement in shelf-life. Biochemical analysis revealed that PAW-treated chillies either maintain or show enhancement in nutritional attributes such as soluble sugar, protein, and ascorbic acid concentrations. Phenol concentration (antioxidant activity) remained stable across treatments. Overall, the study underscores the positive impact of PAW treatment on preserving the membrane integrity, antimicrobial resistance, sensory quality, and nutritional attributes of green chillies, making PAW an alternative for extending their shelf life.

Keywords Plasma activated water · Reactive species · Green chillies · Shelf-life · Nutritional analysis

Extended author information available on the last page of the article

Introduction

Post-harvest food spoilage has extensive and interconnected consequences on a global scale, creating a ripple effect of challenges that impact multiple components of our world. The consequences of food spoilage extend to critical areas such as food shortages, economic losses, environmental degradation, and social issues. The scarcity of edible food resulting from spoilage contributes to widespread food shortages, affecting communities and nations alike. Economically, farmers bear significant financial losses as their harvested crops go to waste before reaching consumers, and this financial strain permeates through the entire supply chain, impacting distributors and retailers. Environmentally, the wastage of resources involved in the production, harvesting, and transportation of food exacerbates environmental strain. Additionally, the decomposition of spoiled food in landfills releases methane, contributing to climate change concerns. Socially, the reduction in the availability of nutritious food leads to increased hunger and malnutrition, especially in regions with limited access to diverse diets, exacerbating food insecurity [1–8].

Microbial contamination stands out as a major contributor to post-harvest food spoilage. As per Central Institute of Post-Harvest Engineering & Technology (CIPHET) 2015 report, approximately ~6.5% spoilage of chillies occurs during post-harvest process [9]. To address this issue, various preservation methods have been employed, including low-temperature preservation, thermal processing, irradiation (UV sterilization, γ-rays, X-rays, etc.), biological methods, and chemical preservatives like alcohol, acids, benzoates, etc. Previously, Chitravathi et al. [10] used combination of aqueous ozone and chlorine for the preservation of post-harvest chillies. However, these techniques often suffer from limitations, such as reduced efficacy, dehydration, nutrient loss, compromise in sensory attributes, and economic impracticality, etc [11–19].

In the pursuit of a preservation technique without these drawbacks, our research led us to a technology known as plasma-activated water (PAW). PAW has gained significant attention in the past decade for its eco-friendly nature, lack of post-processing residue, ease of production, portability, and economic viability, etc [11, 14, 15, 20–30].

Plasma, the fourth state of matter, consists of highly energized ions, electrons, and reactive species. When water is exposed to this plasma, it undergoes various chemical and physical transformations, leading to the creation of PAW. PAW finds applications in agriculture, food safety, medical industries (wound healing, selective killing of cancer cells, dentistry, etc.), water treatment, surface cleaning (inactivation of bacteria, fungi, viruses, pests, etc.), etc [31–43].

Previous research has explored the efficacy of PAW in preserving various foods, including grapes, Chinese bayberries, button mushrooms, fresh-cut apples, lemons, Korean rice cake, shrimps, tofu, chicken meat, etc [14, 15, 23, 25, 28–30, 44–47]. The preservation ability of PAW is attributed to the presence of radicals/species such as hydroxyl radicals (·OH), dissolved ozone (O_3), hydrogen peroxide (H_2O_2), peroxynitrite ($ONOO^-$), and reactive nitrogen species like NO_2^- and NO_3^-, providing an acidic environment that limits microbial growth [22, 34, 36, 44, 46–49].

While the potential of PAW in food preservation has been extensively explored, its effectiveness compared to existing market-based food preservative cleaners has not been thoroughly investigated. The present study aims to fill this gap by comparing the post-harvest food preservation efficacy of PAW with four different commercially available food preservative liquids used for washing and cleaning applications in households.

This investigation focuses on green chillies (*Capsicum annuum* L.), specifically Jalapeño and Pusa Jwala varieties, which play a vital role in global cuisines due to their flavor enhancement, versatility (fresh or dry use), rich nutritional content (vitamin A and C, potassium, iron, antioxidants, etc.), metabolism-boosting properties, and medicinal uses. Green chilies are integral to the culinary traditions of various cultures, contributing significantly to the agricultural industry and global trade [1, 2, 50–57].

In this study, PAW with different reactivity levels is prepared, and its effectiveness is determined based on physicochemical properties and dissolved reactive species concentration. The roles of PAW, control (ultrapure milli-Q water used for PAW preparation), and different commercial cleaners are investigated in enhancing the shelf-life of Jalapeño and Pusa Jwala green chillies. The health of the chillies is monitored periodically through various physical (weight loss, firmness, microbial growth, spoilage), sensory (visual, smell, taste, and touch), and biochemical (soluble sugar and protein, ascorbic acid, phenol) parameters.

Materials and Methods

Schematic of PAW Production

The schematic diagram illustrating the production of plasma-activated water (PAW) and the treatment of chillies is presented in Fig. 1. For PAW production, 250 ml of ultrapure milli-Q water (control) was placed in a 600 ml glass beaker. Ultrapure milli-Q water, devoid of dissolved impurities, was chosen to ensure that only the species/radicals formed during plasma-water interaction played a role in food preservation, eliminating interference from foreign substances. PAWs with different reactivity prepared by varying plasma treatment times between 0 and 60 min, and given the names PAW-20 for 20 min plasma exposed water, PAW-40 to 40 min, and PAW-60 to 60 min plasma exposed water.

The device employed for PAW production was a pencil plasma jet (PPJ) [58]. The PPJ setup, based on the principle of dielectric barrier discharge using quartz as dielectric and air as the plasma-forming gas, utilized a high-frequency power supply for a fixed plasma discharge power (~ 5 W). Supplementary material Table S1 and Figures S1-S4 provide detailed information about the PPJ setup, including its geometry, electrical characterization, and identified plasma species/radicals using optical emission spectroscopy [37, 41, 43, 58, 59].

Physicochemical Properties and Reactive Species Concentration of PAW

The physicochemical properties of PAW, including pH, total dissolved solids (TDS), electrical conductivity (EC), and oxidation-reduction potential (ORP), were measured using a pH meter, TDS meter, EC meter, and ORP meter. Details about these measuring devices are provided in supplementary material Table S2. The concentration of reactive species in PAW, such as NO_3^- ions, NO_2^- ions, H_2O_2, and dissolved O_3, was determined using the NO_3^- ions UV screening method, NO_2^- ions Griess reagent, H_2O_2 Titanium (IV) oxysulfate method, and O_3 indigo method, respectively [60]. Details about these reactive species measurement techniques are presented in supplementary material Table S3.

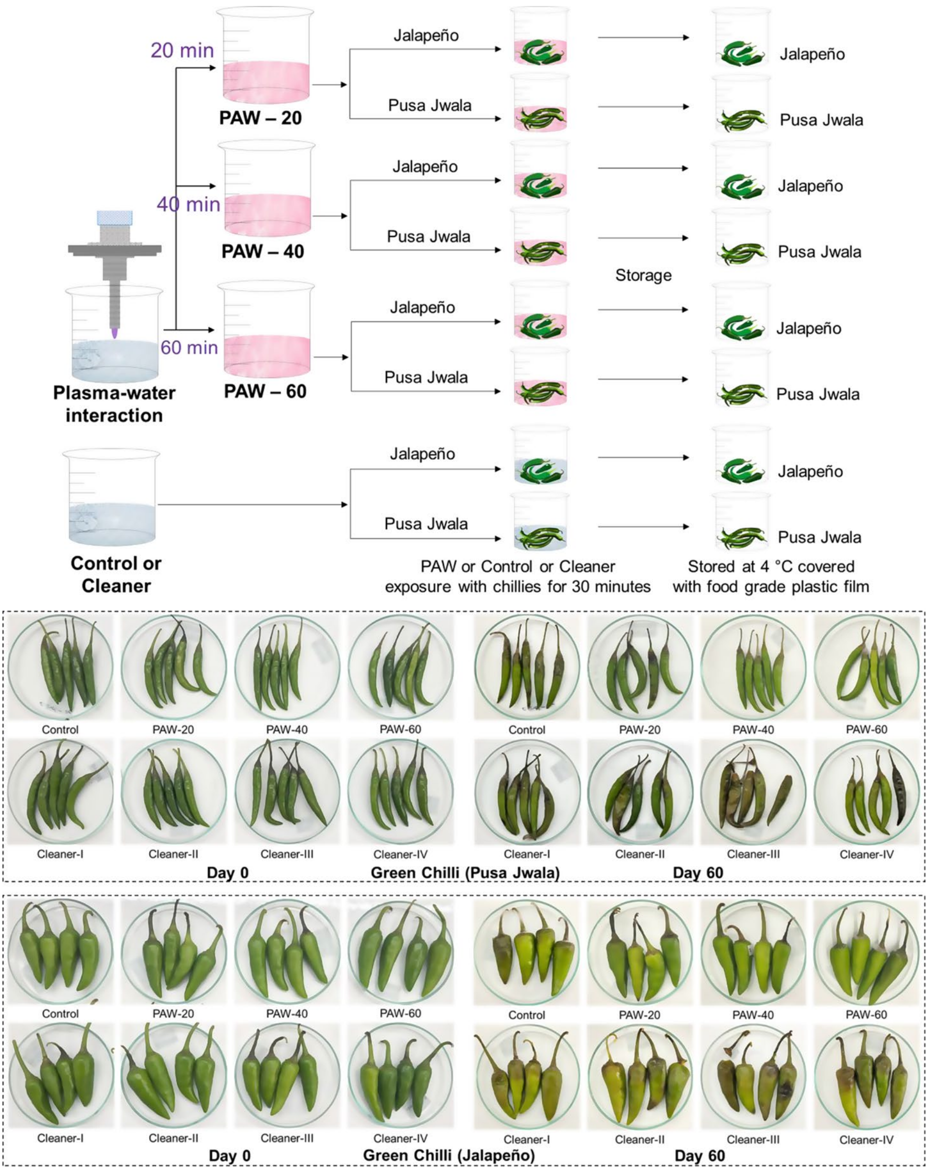

Fig. 1 Schematic of production of plasma activated water (PAW-20, PAW-40 & PAW-60) and exposure of PAW, control, or cleaners to chillies (Jalapeño and Pusa Jwala) with images of chillies over 60-day storage

Treatment of Chillies Using PAW and Cleaners

Chillies Preparation

Green chillies (jalapeño and pusa jwala) (*Capsicum annuum* L.) were procured from the local vegetable market in Gandhinagar, Gujarat, India. All chillies underwent thorough washing with ultrapure milli-Q water to remove settled dirt. Sixteen chilli groups (eight each of jalapeño and pusa jwala) were prepared for different treatments, with each group containing three sets of 20 chillies for replication.

The eight treatment groups were named as follows: Control (ultrapure milli-Q water), PAW-20, PAW-40, PAW-60, Cleaner-I (fruit and vegetable washing liquid containing soapnut, lactic acid, sodium bicarbonate, etc.), Cleaner-II (fruit and vegetable washing liquid containing surfactant, holy basil, Azadirachta indica, *Citrus sinensis* extract, etc.), Cleaner-III (sorbitol ester, etc.), and Cleaner-IV (citrus peel, *Saccharum officinarum* L., essential oil, etc.). All cleaner solutions were prepared following the protocol in the supporting guide.

Treatment of Chillies

Each treatment solution (control, PAW, cleaners) was applied to jalapeño and pusa jwala chillies for 30 min, as depicted in Fig. 1. The treated chillies were incubated at room temperature to eliminate excess moisture and then transferred to sterile containers covered with food-grade plastic film, subsequently stored at 4 °C, as illustrated in Fig. 1. Chillies were examined for various preservation parameters at intervals of 7 days up to 21 days.

Analysis of Chillies

Physical Attributes of Chillies

The physical attributes of chillies after treatment, including weight loss, firmness, microbial load, and spoilage, were analyzed to assess freshness and suitability for consumption. Weight loss and spoilage were calculated using the expressions:

$$Weight\ Loss\ (\%) = \left(\frac{W_{t_0} - W_t}{W_{t_0}}\right) \times\ 100 \tag{1}$$

Where 'W_{t0}' is the initial weight of chilli, 'W_t' is the weight of chilli at time 't'.

$$Spoilage\ (\%) = \left(\frac{N_{t_0} - N_t}{N_{t_0}}\right) \times\ 100 \tag{2}$$

Where 'N_{t0}' is the initial number of unspoiled chilli, 'N_t' is number of unspoiled chilli at time 't'.

The firmness of chillies during storage was measured using a fruit firmness analyser (LUTRON ELECTRONIC, Model Name/Number: FR5120).

Microbial Growth

To analyze microbial growth on chilies, 100 mg of chilies were mixed with 10 ml of sterilized phosphate-buffered saline (PBS). The chili-PBS mixture was vortexed to transfer adhered microbes to PBS. The solution was then centrifuged, and the supernatant was collected for microbial analysis. Microbial growth was calculated using the colony count method. The expression used to calculate normalized microbial growth is given as:

$$Normalized\,microbial\;growth = \frac{(CFU/g)_t}{(CFU/g)_{max}} \tag{3}$$

Where $(CFU/g)_t$ is the microbial count at time 't' and $(CFU/g)_{max}$ is the maximum microbial growth over storage period.

CIELAB Color Analysis of Chillies

The hue of chillies was assessed during storage following treatment with PAW, control, and cleaners, using a colorimeter (NR100 model). The recorded color values were expressed as CIE L×a×b*, where 'L*' represents lightness, 'a*' signifies the color spectrum between red and green, and 'b*' represents the color range between blue and yellow. Chromatic aberration index (croma) (C_{ab}*) and perceived color were computed as per the following expressions:

$$C^*_{ab} = \sqrt{(a^2 + b^2)} \tag{4}$$

C_{ab}* providing a visual indication of color purity. The perceived color was derived from the hue angle, expression given as:

$$h^*_{ab}(^\circ\,) = \tan^{-1}\left(\frac{b}{a}\right) \tag{5}$$

The difference in color (ΔE_{ab}*) of chillies over storage compared to fresh chillies given by the following expression:

$$C^*_{ab} = \sqrt{(\varDelta\,L^*)^2 + (\varDelta\,a^*)^2 + (\varDelta\,b^*)^2} \tag{6}$$

Sensory Analysis

Throughout the storage period, sensory attributes of chillies following control, PAW, and cleaners' treatments were evaluated, including factors like smell, taste, visual appeal, and touch (snap test and wrinkles). A panel of 15 individuals, including both males and females aged 25 to 55, provided ratings on a scale ranging from 1 to 9. The scale used was as follows: 1 (very poor), 3 (poor), 5 (satisfactory), 7 (good), and 9 (excellent).

Biochemical Analysis of Chillies

The analysis of soluble sugars in chilies was conducted using the anthrone reagent. For this analysis, 100 mg of chilies were introduced into 2 ml of warm 80% (v/v) aqueous ethanol. The chilies in this solution underwent a 10-minute heat treatment in a boiling water bath to facilitate sugar extraction. Afterward, the mixture was centrifuged at 12,000 rpm for 5 min, yielding a collected supernatant. Subsequently, 0.2% freshly prepared ice-cold anthrone reagent was added to the supernatant, and the solution was incubated in a boiling water bath for 10 min before being placed on ice. The absorbance of the cooled solution was measured at 590 nm [61]. The quantification of extracted sugar utilized a standard sugar curve prepared using different concentrations of dextrose.

The protein concentration in chilies was determined using the method outlined by Lowry et al. [62]. Chilies (100 mg) were transformed into dry powder using liquid nitrogen (LN2) and a mortar pestle. A protein extraction buffer (lysis buffer) was prepared with 10% trichloroacetic acid (TCA) and 0.07% β-Mercaptoethanol (β-ME) in acetone. The chilies were ground in 2 ml of extraction buffer using an ice-cooled mortar and pestle. The resulting mixture underwent centrifugation at 12,000 rpm for 10 min, and the supernatant was collected. To the collected supernatant, 0.1 ml of 2 N NaOH was added, followed by hydrolysis in a boiling water bath for 10 min. The sample was then cooled to room temperature (25 °C), and 1 ml of a freshly prepared complex reagent (2% Na_2CO_3, 1% $CuSO_4.5H_2O$, 2% $KNaC_4H_4O_6.4H_2O$ in the ratio 100:1:1) was added. The solution was incubated for 10 min at room temperature, and 0.1 ml of Folin reagent was added and left for 30 to 60 min. Finally, the absorbance of the solution was measured at 750 nm. Various concentrations of bovine serum albumin were used to create a standard curve for the protein solution.

To assess the ascorbic acid concentration in chilies, 10% trichloroacetic acid (TCA) was used for the extraction of ascorbic acid from 100 mg chili (dry powder). Centrifuge the mixture and collect the supernatant for ascorbic acid analysis. A 0.1 ml of the supernatant was combined with 0.4 ml of 10% trichloroacetic acid (TCA) and agitated at 3000 rpm for 5 min. The solution was then diluted with 1 ml of ultrapure milli-Q water, and 0.1 ml of freshly prepared Folin reagent was introduced. After shaking for 10 min, the solution underwent an additional 10-minute incubation period before measuring absorbance at 760 nm [63]. The standard curve was established using varying concentrations of ascorbic acid.

Phenol was extracted from 100 mg dry chili powder using a phenol extraction buffer. The solution was centrifuged, and supernatant was collected for phenol analysis. A 0.1 ml of the supernatant was combined with 0.5 ml of a Folin-Ciocalteu reagent (diluted 1:10 with water) and allowed to incubate for 60 s. Following incubation, 0.4 ml of 7.5% sodium carbonate (Na_2CO_3) was added, and the mixture was incubated for an additional 60 min before measuring the absorbance at 765 nm [64]. A standard curve was established using various concentrations of gallic acid to quantify the total soluble phenols present in the chilies.

Data Analysis

Data from the experiments were represented graphically as mean ± standard deviation. The experiments were independently conducted at least three times ($n \geq 3$). Statistical significance ($p < 0.05$) in the results was assessed through analysis of variance (ANOVA), followed by a post-hoc test (Fisher's Least Significant Difference).

Results and Discussion

Images of Green Chillies during Storage

The images of green chillies (jalapeño and pusa jwala) over a 2-month storage period (60 days) following different treatments (control, PAW (PAW-20 to PAW-60), and cleaners (I to IV)) are shown in Fig. 1. Even after 60 days of storage, PAW-washed chillies exhibit significantly better quality compared to control and different cleaners. While PAW-washed chillies show visible microbial growth after 60 days, the control and cleaner-treated chillies are entirely spoiled. Thus, PAW treatment outperformed both the control and cleaners in enhancing the shelf-life of green chillies (jalapeño and pusa jwala).

Physicochemical Properties and Reactive Species Concentration of PAW

The changes in water properties with increasing plasma treatment time are illustrated in Fig. 2. Prolonged treatment renders the water more acidic, demonstrated by a decrease in pH from 6.9 to 5.1 after 60 min of plasma treatment (as shown in Fig. 2 (a)). Simultaneously, it elevates the oxidizing potential from 240 mV to 345 mV, respectively (as shown in Fig. 2 (a)). The decrease in pH and increase in oxidizing potential signify the dissolution of reactive species in water during plasma-liquid interaction. Various reactive species dissolved in water as a result of plasma interaction with water are shown in Fig. 2 (c, d). Since air was used as a plasma-forming gas, the dissolved reactive species in water are predominantly

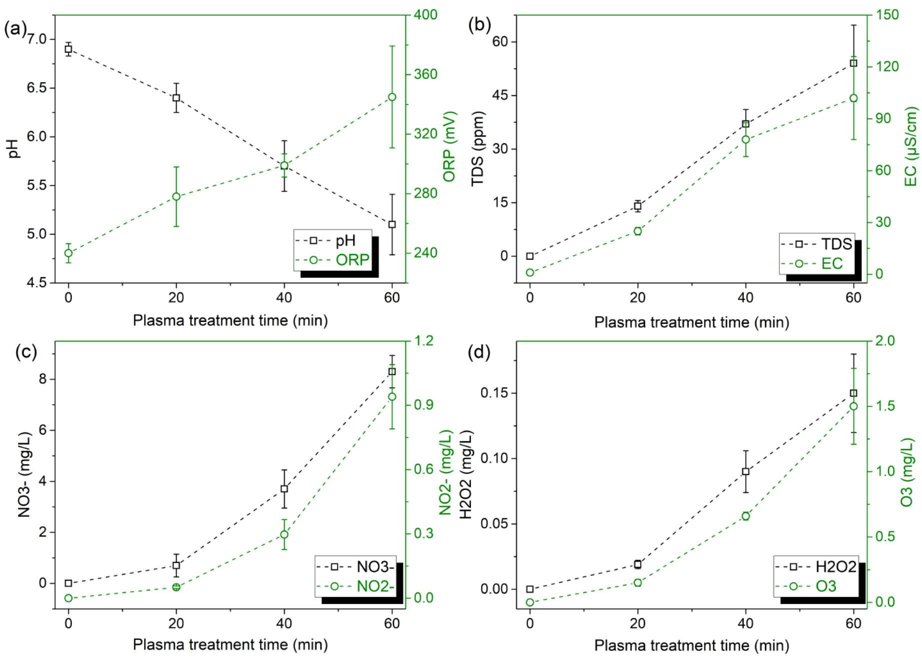

Fig. 2 Change in (**a**, **b**) physicochemical properties (pH, oxidation-reduction potential (ORP)) and (**c**, **d**) reactive species concentration (NO_3^- ions, NO_2^- ions, H_2O_2, and Dissolved O_3) of plasma activated water with increasing plasma treatment time

inorganic. These dissolved species enhance the total dissolved solids (TDS) and electrical conductivity (EC) of water, reaching 54 ppm and 102 µS cm^{-1}, respectively, after 60 min of plasma treatment (Fig. 2 (b)). Past literature supports the findings of the present work, emphasizing a decrease in pH, an increase in ORP, and EC of PAW with increasing plasma treatment time [21, 24, 49, 65–70].

Air plasma contains species like N_2, N_2^+, N, NOx, O, O_2^+, O_3, OH, H, e^-, etc. These species and radicals, when in contact with water, form species like H^+ ions, NO_2^- ions, NO_3^- ions, $ONOO^-$ ions, OH, H_2O_2, dissolved O_3, etc. The proposed reactions for the formation of these RONS in PAW is presented in supplementary material Table S4. The variation in the concentration of NO_2^- and NO_3^- ions with increasing plasma treatment time is shown in Fig. 2 (c). A monotonic increase in the concentration of NO_2^- and NO_3^- ions is observed with treatment time, with a maximum concentration of 0.94 mg L^{-1} and 8.3 mg L^{-1} at the end of the treatment duration. These NO_2^- and NO_3^- ions, along with H^+ in PAW, form nitrous and nitric acid in water, resulting in decreased pH and the observed acidic nature of PAW (Fig. 2 (a)). Furthermore, increasing plasma treatment time enhances the H_2O_2 and dissolved O_3 in PAW, as depicted in Fig. 2 (d). These H_2O_2 and dissolved O_3 play a significant role in the antimicrobial properties of PAW. As all the dissolved reactive species are oxidizing in nature, the oxidizing potential of water also increases, as observed in the results of ORP (Fig. 2 (a)). This increase in RONS concentration of PAW with increasing treatment time aligns with the reported works of Jin et al. [34], Pan et al. [49], and Than et al. [32], etc.

Physical Attributes of Chillies after PAW and Cleaners' Treatment

Weight Loss

Figure 3 (a, b) illustrates the weight loss of chillies (jalapeño and pusa jwala) over the storage duration. PAW-treated chillies (jalapeño and pusa jwala) exhibited a lower percentage of weight loss compared to control and cleaners' treated chillies. After 21 days of storage, cleaner-I treated jalapeño showed the highest percentage weight loss (15.04%), and cleaner-II treated pusa jwala exhibited the highest weight loss (14.58%). Meanwhile, PAW-treated jalapeño and pusa jwala demonstrated weight loss ranging between 5.96 and 6.79% and 7.49–8.38%, respectively. The reduced weight loss in PAW-treated chillies indicates better membrane integrity, aiding in moisture retention during storage. Liu et al. [28] and Zhao et al. [71] reported weight loss in food products (fresh-cut apples and button mushrooms) after PAW exposure. Liu et al. [28] observed a 2–4% weight loss in fresh-cut apples after PAW treatment, compared to approximately 2.3% in the control group over a storage period of 12 days. Zhao et al. [71] found that mushrooms treated with PAW experienced less than 3% weight loss, whereas those treated with control or water showed a weight loss between 3 and 5% over the same period. These findings indicate that PAW treatment generally results in lower or insignificant weight loss compared to control treatments, aligning with the present study's results, which showed reduced weight loss in PAW-treated chilies compared to control treated samples.

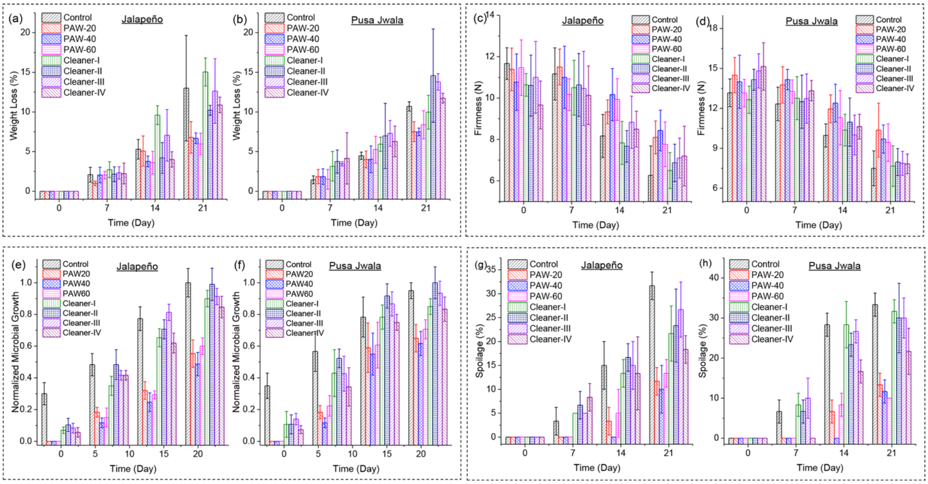

Fig. 3 Change in physical attributes ((**a**, **b**) weight loss, (**c**, **d**) firmness, (**e**, **f**) microbial growth, and (**g**, **h**) spoilage) of green chillies (jalapeño and pusa jwala) over storage after treatment with control (Ultrapure Milli-Q Water), PAW (PAW-20, PAW-40, and PAW-60), and cleaners (Cleaner-I to Cleaner-IV)

Firmness

Membrane integrity evaluation of PAW-treated chillies based on firmness (Fig. 3 (c, d)) revealed higher firmness in PAW-treated chillies (jalapeño and pusa jwala) compared to control and cleaners' treated chillies. PAW-treated jalapeño exhibited firmness 7.78–34.44% higher than control and cleaners' treated jalapeño after 21 days. Similarly, PAW-treated pusa jwala showed 30.11–38.27% higher firmness compared to control and cleaners' treatment. The higher firmness and lower weight loss of PAW-treated jalapeño and pusa jwala indicate the significant role of PAW treatment in preserving the membrane integrity of chillies. Past studies of Liu et al. [28], Zhao et al. [71], Ma et al. [22], Xiang et al. [15], and Zheng et al. [25], etc. also emphasis on the fact that the PAW treatment did not affect the firmness of food. Similar to the present investigation, Zheng et al. [25], Liu et al. [28] and Zhao et al. [71] also reported higher firmness in PAW-treated food products compared to the control over the storage duration. Additionally, Ma et al. [22] and Xiang et al. [15] observed no significant changes in firmness between PAW-treated and control samples during the storage period.

Spoilage

Chillies spoilage after PAW and cleaners' treatment over the storage duration is shown in Fig. 3 (g, h). PAW-20 treated chillies did not show any spoilage until the 14th day of storage. Meanwhile, control and cleaners' treated chillies showed more than 10% spoilage. Chillies spoilage after control and cleaners' treatment increased by 18–32% for jalapeño and 21–34% for pusa jwala after 21 days of storage. However, the chillies (jalapeño and pusa jwala) spoilage after PAW treatment was less than 13.5% for a similar storage duration. Previously, Ma et al. [22] also reported the positive impact of PAW on reducing strawberries spoilage. The strawberries treated with PAW showed no visible bacterial or fungal growth, while grey mold was observed on the strawberries treated with water and in the control group.

Microbial Growth

Microbial infection is a major cause of post-harvest fruits and vegetables spoilage. Figure 3 (e, f) displays the normalized microbial growth on chillies after PAW and cleaners' treatment over storage. PAW demonstrated significantly better microbial growth resistance compared to control and cleaners. The oxidizing species like dissolved O_3 and H_2O_2, along with the acidic nature of PAW (Fig. 2), make it a suitable antimicrobial agent, resulting in better antimicrobial activities on chillies compared to other cleaners. After 21 days of chillies storage, PAW-treated jalapeño showed 29.14–51.33% lower microbial growth compared to control and cleaners' treated chillies. For pusa jwala, this microbial reduction ranged between 15.28 and 38.34% compared to control and cleaners' treated chillies. Previous literature also showed PAW significantly reduces microbial growth on food products, as reported by Liu et al. [28], Kang et al. [29], and Liao et al. [30], etc. Ke et al. [72] reported a reduction of 0.37 and 2.23 log CFU mL^{-1} in Shewanella putrefaciens when exposed to PAW-150 and PAW-300, respectively. *Shewanella putrefaciens*, a common spoilage bacterium in marine and freshwater environments, showed percentage reductions of 4.47% and 28.77% under these conditions. Similarly, Nasiru et al. [73] demonstrated reductions in various pathogenic microbes associated with eggs, including *Staphylococcus aureus*, *Salmonella Enteritidis*, and *Campylobacter jejuni*. The maximum reductions in microbial load for these pathogens when exposed to PAW were 48.34%, 28.87%, and 31.28%, respectively. Present findings, along with those of Ke et al. [72] and Nasiru et al. [73], highlight a consistent reduction in microbial load on food products, thereby proving the antimicrobial efficacy of PAW.

In summary, PAW treatment preserves the membrane integrity of chillies, as shown by weight loss and firmness analysis, inhibits the growth of food spoilage microbes, and significantly reduces chillies spoilage over storage.

CIELAB Color Analysis

Food color, including that of fruits and vegetables, is a crucial parameter for assessing the freshness of food. Figure 4 illustrates the variation in CIELAB color parameters of chillies over storage after PAW and cleaners' treatment. Figure 4 (b) highlights the retention of greenness in PAW-treated jalapeño even after 21 days of storage, in contrast to control and cleaners. The reduction in greenness of control and cleaners' treated jalapeño is evident through the decrease in '-a*' value after 21 days (Day 21: a* value (PAW-treated jalapeño): -16.5 to -17.5; a* value (control and cleaners' treated jalapeño): -13.28 to -14.96). Similarly, PAW-treated pusa jwala exhibited higher greenness (a* value: -12.01 to -12.61) compared to cleaners treated pusa jwala (a* value: -9.63 to -10.97) over the storage duration (Fig. 4 (h)).

The observed color difference (ΔE_{ab}*) over the 21-day storage period in PAW-40 (ΔEab*: 3.047) and PAW-60 (ΔEab*: 3.084) treated jalapeño was significantly lower compared to other jalapeño treatments (ΔE_{ab}*: 4.523 to 14.56) as shown in Fig. 4 (d). This preservation of jalapeño color over the storage duration indicates that PAW indeed contributes to retaining the freshness of jalapeño. A similar trend is observed in PAW-treated pusa jwala compared to control and cleaners' treated pusa jwala, as shown in Fig. 4 (j). After the 21-day

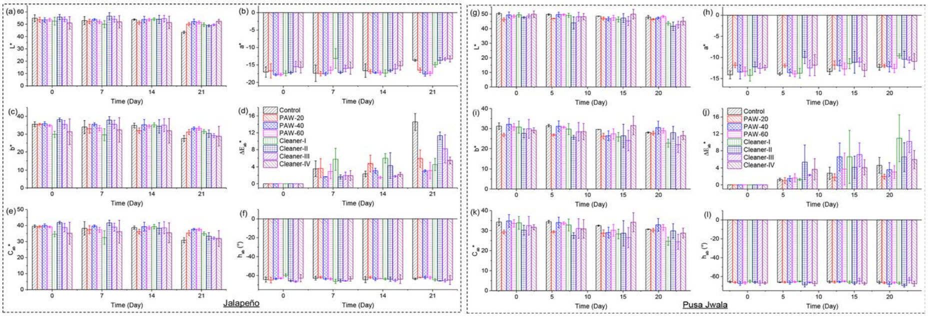

Fig. 4 Change in CIELAB color ((**a**, **g**) L*, (**b**, **h**) a*, (**c**, **i**) b*, (**d**, **j**) ΔE_{ab}*, (**e**, **k**) C_{ab}*, and (**f**, **l**) h_{ab}) of green chillies (jalapeño and pusa jwala) over storage after treatment with control (ultrapure milli-Q water), PAW (PAW-20, PAW-40, and PAW-60), and cleaners (cleaner-I to cleaner-IV)

storage duration, the change in color in PAW-treated pusa jwala was less than 3.6 (ΔEab*), while the ΔEab* value of control and cleaners' treated pusa jwala ranged from 4.6 to 11.0, respectively. The results of chroma (C_{ab}*) and hue angle (h_{ab}) analysis (Fig. 4 (e, f, k, l)) further support that PAW treatment with chillies retains the color or remains closer to the initial color of fresh chillies. In line with the present investigation, the retention of food surface color after PAW treatment has also been reported by Lotfy et al. [74] and Zhao et al. [71] Lotfy et al. [74] found that a low dose of plasma or shorter plasma treatment time during PAW production helps retain the surface color of beef. Zhao et al. [71] observed a lower browning index (BI) in PAW-treated button mushrooms compared to the control. Additionally, Ma et al. [22] and Tan et al. [75] reported no significant change in the color of strawberries and navel oranges treated with either PAW or the control.

Sensory Evaluation

Sensory analysis of food is a fundamental assessment to determine freshness and suitability for consumption. The sensory evaluation of chillies after PAW and cleaners' treatment in this study includes visual inspection, smell, taste, and touch.

Results of visual inspection of PAW, control, and cleaners' treated jalapeño and pusa jwala over storage are shown in Fig. 5 (a, e). Visual inspection clearly demonstrates the efficacy of PAW in enhancing the shelf life of jalapeño and pusa jwala compared to control and cleaners. PAW-treated jalapeño, especially PAW-40 (after 21 days of storage), appears significantly better than control and cleaners' treatment (Fig. 5 (a)). Moreover, all PAW-treated pusa jwala after 21 days of storage appears as fresh as on day 0 (Fig. 5 (b)). After 21 days of storage, control and cleaners treated pusa jwala showed degradation in visual inspection, with cleaner-II treated pusa jwala being beyond visual acceptance.

Similar to visual inspection, PAW-treated jalapeño and pusa jwala retained the smell and taste equivalent to fresh chillies, as shown in Fig. 5 (b, f). Among these, the PAW-40 treatment with chillies yielded better results compared to PAW-20 and PAW-60 treatments. This emphasizes the importance of optimizing plasma treatment time for PAW production, as the reactivity of PAW depends on the concentration of various reactive oxygen and nitrogen species (RONS). Since the concentration of RONS relies on plasma treatment time, the reactivity of PAW is intricately linked to plasma-water interaction time (Fig. 2). Low plasma

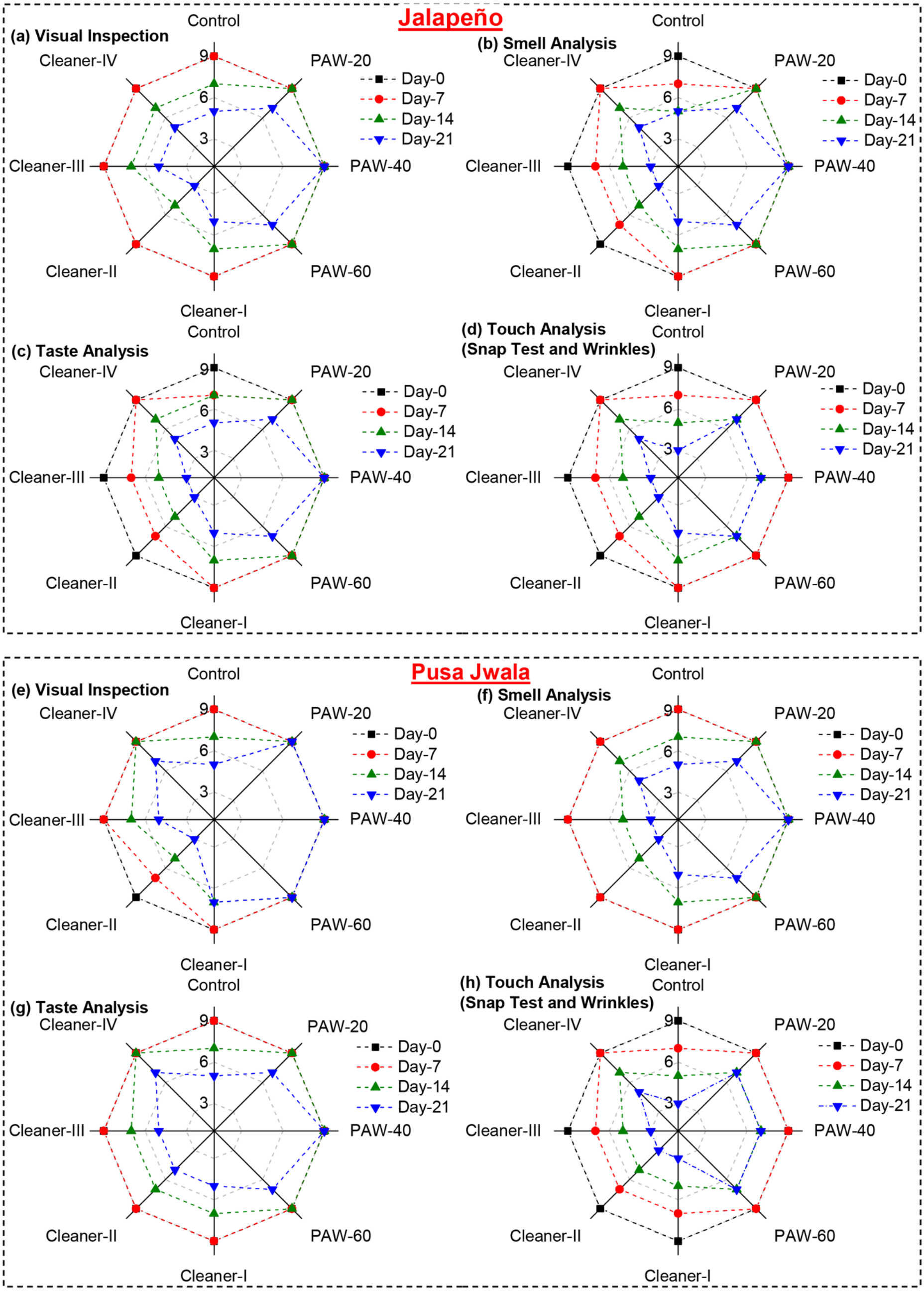

Fig. 5 Sensory analysis ((**a**, **e**) visual inspection, (**b**, **f**) smell Analysis, (**c**, **g**) taste analysis, and (**d**, **h**) snap test and wrinkles (touch analysis)) of green chillies (jalapeño and pusa jwala) over storage after treatment with control (ultrapure milli-Q water), PAW (PAW-20, PAW-40, and PAW-60), and cleaners (Cleaner-I to Cleaner-IV)

interaction time with water resulted in the production of a less reactive PAW, which may not significantly enhance the shelf-life of food. Conversely, highly reactive PAW may oxidize the food itself, eventually may lead to unacceptable food quality. The superior performance of PAW-40 over PAW-20 and PAW-60 may be the proposed explanation based on above discussion. Among the cleaners, cleaner-I and cleaner-IV treated jalapeño and pusa jwala showed relatively better smell and taste compared to cleaner-II and cleaner-III.

The touch analysis involves observing the wrinkles on the chillies' surface, and the snap test entails holding the chilli near the stem and attempting to bend it. Fresh chillies generally have a crisp texture and should produce a snapping sound when broken. If they bend without snapping, they might be less fresh. The PAW-treated chillies (jalapeño and pusa jwala) performed significantly better in touch analysis compared to control and cleaners' treated chillies, as shown in Fig. 5 (d, h). Even after 21 days of storage, no significant observable wrinkles were found on PAW-treated chillies' surfaces, and they produced a snapping sound when broken.

The sensory evaluation clearly demonstrated the superior performance of PAW treatment in enhancing the shelf life of chillies compared to control and different cleaners' treatment. The earlier work of Liu et al. [28] demonstrated that the overall sensory score of fresh-cut apples treated with the control fell below the market-acceptable limit by day 6, whereas this limit was extended to day 12 for PAW-treated apples. These findings support the results of the present investigation, indicating that PAW treatment can be used to prolong the market-acceptable shelf life of food products.

Biochemical Analysis

Figure 6 presents various biochemical analyses of chillies after treatment with PAW and cleaners at day 0 and 21. The sugar concentration of chillies marginally decreased over the storage duration (Fig. 6 (a, b)). After 21 days of chillies storage, the soluble sugar concentration of PAW-treated jalapeño and pusa jwala was higher compared to control and cleaners. The maximum sugar concentration was observed in PAW-20 treated jalapeño and pusa jwala, being 32.76% and 26.67% higher compared to control (ultrapure milli-Q water) and 17.66% and 38.33% higher compared to the maximum sugar concentration in cleaner-treated chillies. However, the sugar concentration of PAW-treated jalapeño was not statistically significant ($p > 0.05$) compared to cleaner-I and cleaner-IV. Additionally, the sugar concentration of PAW-treated pusa jwala was not statistically significant ($p > 0.05$) compared to control. This insignificant variation in sugar concentration was also observed by Xiang et al. [15] in their investigation of the effect of PAW on grapes.

The variation in protein and ascorbic acid concentration of jalapeño and pusa jwala after PAW and cleaners' treatment over storage is shown in Fig. 6 (c-f). PAW-treated jalapeño and pusa jwala showed higher protein and ascorbic acid concentrations compared to control and cleaners. The maximum protein and ascorbic acid concentrations in chillies after the storage duration were observed in PAW-40 treatment. The observed maximum values of protein and ascorbic acid in jalapeño and pusa jwala were 3.28 mg and 3.98 mg of protein per gm of chillies and 0.82 mg and 0.91 mg ascorbic acid per gm of chillies, respectively. However, this enhanced protein and ascorbic acid concentration of jalapeño after PAW treatment were not statistically significantly ($p > 0.05$) different from cleaner-I and cleaner-IV, except for PAW-40 (Fig. 6 (c, e)). The ascorbic acid concentration of PAW-treated pusa jwala was sig-

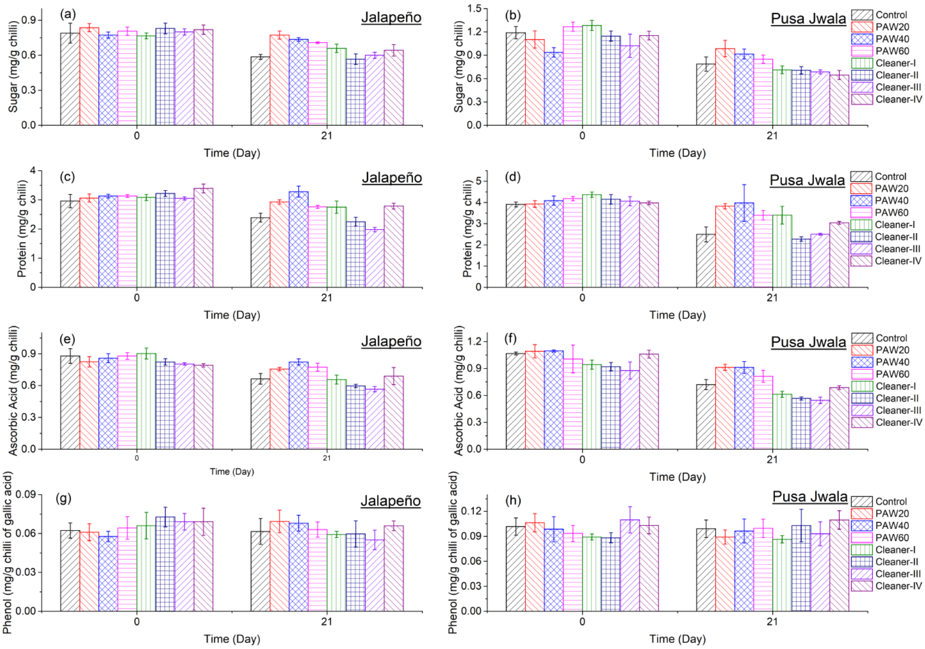

Fig. 6 Biochemical (nutritional and antioxidant) analysis ((**a**, **b**) soluble sugar, (**c**, **d**) soluble protein, (**e**, **f**) ascorbic acid, and (**g**, **h**) soluble phenol) of green chillies (jalapeño and pusa jwala) over storage after treatment with control (ultrapure milli-Q water), PAW (PAW-20, PAW-40, and PAW-60), and cleaners (Cleaner-I to Cleaner-IV)

nificantly higher compared to other cleaners treated pusa jwala (Fig. 6 (f)). A past report by Royintarat et al. [26] showed that PAW treatment did not change the protein concentration of chicken meat. Moreover, PAW treatment did not alter the ascorbic acid concentration of grapes, as shown in the work of Xiang et al. [15].

This elevated level of sugar, protein, and ascorbic acid in PAW-treated chillies signifies the better health of chillies. Therefore, it can be concluded that PAW treatment enhances the shelf life of green chillies while retaining their physical and biochemical attributes. The phenol concentration (antioxidant activity) of chillies treated with PAW, control, and cleaners did not show any significant variation over storage as shown in Fig. 6 (g, h). Hence, PAW treatment with chillies did not alter the antioxidant activity of chillies.

The results discussed in this study emphasize the efficacy of plasma-activated water (PAW) in preserving green chillies (jalapeño and pusa jwala) during a 21-day storage period, comparing it with control and various cleaners. Visual analysis reveals that PAW-treated chillies maintain a superior appearance even after 60 days, exhibiting significantly lower spoilage compared to control and cleaners. Physicochemical properties of PAW, including acidity and oxidizing potential, increase with plasma treatment time, influencing the concentration of reactive species. PAW demonstrates antimicrobial properties, effectively reducing microbial growth and spoilage on chillies over the storage period.

Physical attributes such as weight loss and firmness are evaluated, with PAW-treated chillies exhibiting lower weight loss and higher firmness, indicating better membrane integrity and moisture retention. Microbial resistance is notably higher in PAW-treated chillies compared to control and cleaners. CIELAB color analysis reveals that PAW-treated chillies retain greenness and color freshness, outperforming control and cleaners. Sensory evaluation, including visual inspection, smell, taste, and touch, consistently favors PAW-treated chillies, emphasizing their superior shelf-life extension.

Biochemical analysis shows that PAW-treated chillies maintain or enhance nutritional attributes such as soluble sugar, protein, and ascorbic acid concentrations. Phenol concentration (antioxidant activity) remains stable across treatments. Overall, the study underscores the positive impact of PAW treatment on preserving the membrane integrity, antimicrobial resistance, sensory quality, and nutritional attributes of green chillies, making it a promising method for extending their shelf life.

Conclusion

The study highlights the effectiveness of plasma-activated water (PAW) in enhancing the shelf life of green chillies (jalapeño and pusa jwala) over a 21-day storage period. PAW-treated chillies consistently outperform control and various cleaners in terms of visual appearance, microbial growth inhibition, and spoilage reduction. Physicochemical analysis reveals increased acidity and oxidizing potential of PAW, contributes to its antimicrobial properties. PAW-treated chillies show improved membrane integrity, lower weight loss, and higher firmness. CIELAB color analysis confirms the retention of greenness and overall color freshness. Sensory evaluation consistently favors PAW-treated chillies, and biochemical analysis indicates maintained or enhanced nutritional attributes.

Overall, the study underscores the significant potential of PAW for prolonging the shelf life of green chillies while preserving quality and nutritional value. In conclusion, PAW has the potential to replace conventional cleaners which are used for food preservation in households, restaurants, and various food-related industries.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s11090-024-10509-0.

Author Contributions All authors were involved in the conceptualization and design of the study. V.R. handled material preparation, data collection, and analysis. V.R. wrote the initial draft of the manuscript and P.S., A.P.V., and S.K.N. performed the review and editing of the manuscript. The final manuscript was reviewed and approved by all authors.

Funding Open access funding provided by Department of Atomic Energy.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Ethical Approval Not applicable.

Competing Interests The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

1. Eelager MP, Masti SP, Chougale RB, Hiremani VD, Narasgoudar SS, Dalbanjan NP et al (2023) Evaluation of mechanical, antimicrobial, and antioxidant properties of vanillic acid induced chitosan/poly (vinyl alcohol) active films to prolong the shelf life of green Chilli. Int J Biol Macromol 232:123499
2. Singh R, Giri S, Kotwaliwale N (2014) Shelf-life enhancement of green bell pepper (Capsicum annuum L.) under active modified atmosphere storage. Food Packaging Shelf Life 1(2):101–112
3. Ali A, Xia C, Mahmood I, Faisal M (2021) Economic and environmental consequences' of postharvest loss across food supply chain in the developing countries. J Clean Prod 323:129146
4. Kumar D, Kalita P (2017) Reducing postharvest losses during storage of grain crops to strengthen food security in developing countries. Foods 6(1):8
5. Kiaya V (2014) Post-harvest losses and strategies to reduce them. Technical Paper on Postharvest losses. Action Contre La Faim (ACF) 25:1–25
6. Gangwar RK, Tyagi S, Kumar V, Singh K, Singh G (2014) Food production and post harvest losses of food grains in India. Food Sci Qual Manage 31:48–52
7. Negi S, Anand N (2017) Post-harvest losses and Wastage in Indian Fresh Agro Supply Chain Industry: a challenge. IUP J Supply Chain Manage 14(2)
8. AlQahtani S, Khan S, Ahmed E, AlQahtani AEA (2024) Sustainable Management of Food Waste in Saudi Arabia. Food and Nutrition Security in the Kingdom of Saudi Arabia. Macroeconomic Policy and Its Implication on Food and Nutrition Security, vol 2. Springer, pp 215–239
9. CIPHET (2015) Report on Assessment of quantitative Harvest and Post-harvest losses. ICAR - Central Institute of Post-Harvest Engineering & Technolog
10. Chitravathi K, Chauhan O, Raju P, Madhukar N (2015) Efficacy of aqueous ozone and chlorine in combination with passive modified atmosphere packaging on the postharvest shelf-life extension of green chillies (Capsicum annuum L). Food Bioprocess Technol 8:1386–1392
11. Zhao N, Ge L, Huang Y, Wang Y, Wang Y, Lai H et al (2020) Impact of cold plasma processing on quality parameters of packaged fermented vegetable (radish paocai) in comparison with pasteurization processing: insight into safety and storage stability of products. Innovative Food Sci Emerg Technol 60:102300
12. Saravanakumar K, Hu X, Chelliah R, Oh D-H, Kathiresan K, Wang M-H (2020) Biogenic silver nanoparticles-polyvinylpyrrolidone based glycerosomes coating to expand the shelf life of fresh-cut bell pepper (Capsicum annuum L. var. Grossum (L.) Sendt). Postharvest Biol Technol 160:111039
13. Salama HE, Aziz MSA (2020) Optimized carboxymethyl cellulose and guanidinylated chitosan enriched with titanium oxide nanoparticles of improved UV-barrier properties for the active packaging of green bell pepper. Int J Biol Macromol 165:1187–1197
14. Suganya A, Shanmugvelayutham G, Hidalgo-Carrillo J (2018) Plasma surface modified polystyrene and grafted with chitosan coating for improving the shelf lifetime of postharvest grapes. Plasma Chem Plasma Process 38:1151–1168
15. Xiang Q, Zhang R, Fan L, Ma Y, Wu D, Li K et al (2020) Microbial inactivation and quality of grapes treated by plasma-activated water combined with mild heat. Lwt 126:109336
16. Kondratowicz J, Matusevičius P (2002) Use of low temperatures for food preservation. Veterinarija Ir Zootechnika 17(39):88–92
17. Zhang S, Luo L, Sun X, Ma A (2021) Bioactive peptides: a promising alternative to chemical preservatives for food preservation. J Agric Food Chem 69(42):12369–12384
18. Loaharanu P, Ahmed M (1991) Advantages and disadvantages of the use of irradiation for food preservation. J Agric Environ Ethics 4:14–30
19. Pereira RN, Vicente AA (2010) Environmental impact of novel thermal and non-thermal technologies in food processing. Food Res Int 43(7):1936–1943
20. Rathore V, Patel D, Butani S, Nema SK (2021) Investigation of physicochemical properties of plasma activated water and its bactericidal efficacy. Plasma Chem Plasma Process 41:871–902
21. Zhang Q, Liang Y, Feng H, Ma R, Tian Y, Zhang J et al (2013) A study of oxidative stress induced by non-thermal plasma-activated water for bacterial damage. Appl Phys Lett 102(20)

22. Ma R, Wang G, Tian Y, Wang K, Zhang J, Fang J (2015) Non-thermal plasma-activated water inactivation of food-borne pathogen on fresh produce. J Hazard Mater 300:643–651
23. Rathore V, Nema SK (2023) Enhancement of shelf life of Citrus Limon L.(Lemon) using plasma activated water. Plasma Chem Plasma Process 43:1109–1129
24. Thirumdas R, Kothakota A, Annapure U, Siliveru K, Blundell R, Gatt R et al (2018) Plasma activated water (PAW): Chemistry, physico-chemical properties, applications in food and agriculture. Trends Food Sci Technol 77:21–31
25. Zheng Y, Wu S, Dang J, Wang S, Liu Z, Fang J et al (2019) Reduction of phoxim pesticide residues from grapes by atmospheric pressure non-thermal air plasma activated water. J Hazard Mater 377:98–105
26. Royintarat T, Choi EH, Boonyawan D, Seesuriyachan P, Wattanutchariya W (2020) Chemical-free and synergistic interaction of ultrasound combined with plasma-activated water (PAW) to enhance microbial inactivation in chicken meat and skin. Sci Rep 10(1):1559
27. Rathore V, Patel D, Shah N, Butani S, Pansuriya H, Nema SK (2021) Inactivation of Candida albicans and lemon (Citrus limon) spoilage fungi using plasma activated water. Plasma Chem Plasma Process 41(5):1397–1414
28. Liu C, Chen C, Jiang A, Sun X, Guan Q, Hu W (2020) Effects of plasma-activated water on microbial growth and storage quality of fresh-cut apple. Innovative Food Sci Emerg Technol 59:102256
29. Kang C, Xiang Q, Zhao D, Wang W, Niu L, Bai Y (2019) Inactivation of Pseudomonas deceptionensis CM2 on chicken breasts using plasma-activated water. J Food Sci Technol 56:4938–4945
30. Liao X, Su Y, Liu D, Chen S, Hu Y, Ye X et al (2018) Application of atmospheric cold plasma-activated water (PAW) ice for preservation of shrimps (Metapenaeus ensis). Food Control 94:307–314
31. Rathore V, Nema SK (2023) Selective generation of reactive oxygen species in plasma-activated water using CO2 plasma. J Vacuum Sci Technol A 41(4)
32. Than HAQ, Pham TH, Nguyen DKV, Pham TH, Khacef A (2022) Non-thermal plasma activated water for increasing germination and plant growth of Lactuca sativa L. Plasma Chem Plasma Process 1–17
33. Rathore V, Jamnapara NI, Nema SK (2023) Enhancing the physicochemical properties and reactive species concentration of plasma activated water using an air bubble diffuser. Phys Lett A 482:129035
34. Jin YS, Cho C, Kim D, Sohn CH, Ha C-s, Han S-T (2020) Mass production of plasma activated water by an atmospheric pressure plasma. Jpn J Appl Phys 59(SH):SHHF05
35. Rathore V, Patil C, Sanghariyat A, Nema SK (2022) Design and development of dielectric barrier discharge setup to form plasma-activated water and optimization of process parameters. Eur Phys J D 76(5):77
36. Guo L, Xu R, Gou L, Liu Z, Zhao Y, Liu D et al (2018) Mechanism of virus inactivation by cold atmospheric-pressure plasma and plasma-activated water. Appl Environ Microbiol 84(17):e00726–e00718
37. Rathore V, Nema SK (2022) A comparative study of dielectric barrier discharge plasma device and plasma jet to generate plasma activated water and post-discharge trapping of reactive species. Phys Plasmas 29(3)
38. Subramanian PG, Jain A, Shivapuji AM, Sundaresan NR, Dasappa S, Rao L (2020) Plasma-activated water from a dielectric barrier discharge plasma source for the selective treatment of cancer cells. Plasma Processes Polym 17(8):1900260
39. Rathore V, Nema SK (2022) The role of different plasma forming gases on chemical species formed in plasma activated water (PAW) and their effect on its properties. Phys Scr 97(6):065003
40. Milhan NVM, Chiappim W, Sampaio AG, Vegian MRC, Pessoa RS, Koga-Ito CY (2022) Applications of plasma-activated water in dentistry: a review. Int J Mol Sci 23(8):4131
41. Rathore V, Tiwari BS, Nema SK (2022) Treatment of pea seeds with plasma activated water to enhance germination, plant growth, and plant composition. Plasma Chem Plasma Process 1–21
42. Wang S, Xu D, Qi M, Li B, Peng S, Li Q et al (2021) Plasma-activated water promotes wound healing by regulating inflammatory responses. Biophysica 1(3):297–310
43. Rathore V, Nema SK (2023) Investigating the role of plasma-activated water on the growth of freshwater algae Chlorella pyrenoidosa and Chlorella sorokiniana. Plasma Chem Plasma Process 1–25
44. Ma R, Yu S, Tian Y, Wang K, Sun C, Li X et al (2016) Effect of non-thermal plasma-activated water on fruit decay and quality in postharvest Chinese bayberries. Food Bioprocess Technol 9:1825–1834
45. Xu Y, Tian Y, Ma R, Liu Q, Zhang J (2016) Effect of plasma activated water on the postharvest quality of button mushrooms, Agaricus Bisporus. Food Chem 197:436–444
46. Han J-Y, Song W-J, Kang JH, Min SC, Eom S, Hong EJ et al (2020) Effect of cold atmospheric pressure plasma-activated water on the microbial safety of Korean rice cake. LWT 120:108918
47. Frías E, Iglesias Y, Alvarez-Ordóñez A, Prieto M, González-Raurich M, López M (2020) Evaluation of Cold Atmospheric pressure plasma (CAPP) and plasma-activated water (PAW) as alternative non-thermal decontamination technologies for tofu: impact on microbiological, sensorial and functional quality attributes. Food Res Int 129:108859

48. Rathore V, Desai V, Jamnapara NI, Nema SK (2022) Activation of water in the downstream of low-pressure ammonia plasma discharge. Plasma Res Express 4(2):025008
49. Pan J, Li Y, Liu C, Tian Y, Yu S, Wang K et al (2017) Investigation of cold atmospheric plasma-activated water for the dental unit waterline system contamination and safety evaluation in vitro. Plasma Chem Plasma Process 37:1091–1103
50. Samira A, Woldetsadik K, Workneh TS (2013) Postharvest quality and shelf life of some hot pepper varieties. J Food Sci Technol 50:842–855
51. Chitravathi K, Chauhan OP, Kizhakkedath J (2020) Shelf life extension of green chillies (Capsicum annuum L.) using passive modified atmosphere packaging and gamma irradiation. J Food Process Preserv 44(8):e14622
52. Chitravathi K, Chauhan O, Raju P (2014) Postharvest shelf-life extension of green chillies (Capsicum annuum L.) using shellac-based edible surface coatings. Postharvest Biol Technol 92:146–148
53. Cheema A, Padmanabhan P, Amer A, Parry MJ, Lim L-T, Subramanian J et al (2018) Postharvest hexanal vapor treatment delays ripening and enhances shelf life of greenhouse grown sweet bell pepper (Capsicum annum L). Postharvest Biol Technol 136:80–89
54. Panigrahi J, Gheewala B, Patel M, Patel N, Gantait S (2017) Gibberellic acid coating: a novel approach to expand the shelf-life in green Chilli (Capsicum annuum L). Sci Hort 225:581–588
55. Patel N, Gantait S, Panigrahi J (2019) Extension of postharvest shelf-life in green bell pepper (Capsicum annuum L.) using exogenous application of polyamines (spermidine and putrescine). Food Chem 275:681–687
56. Meng X, Zhang M, Adhikari B (2012) Extending shelf-life of fresh-cut green peppers using pressurized argon treatment. Postharvest Biol Technol 71:13–20
57. Maurya VK, Ranjan V, Gothandam KM, Pareek S (2020) Exogenous gibberellic acid treatment extends green Chili shelf life and maintain quality under modified atmosphere packaging. Sci Hort 269:108934
58. Rathore V, Nema SK (2021) Optimization of process parameters to generate plasma activated water and study of physicochemical properties of plasma activated solutions at optimum condition. J Appl Phys 129(8)
59. Rathore V, Pandey A, Patel S, Dave H, Nema SK (2024) Degradation of dyes using reactive species of atmospheric pressure dielectric barrier discharge formed by a pencil plasma jet. Phys Scr 99(3):035602
60. Rathore V, Patil C, Nema SK (2024) Production of large quantity of plasma activated water using multiple plasma device setup. Curr Appl Phys 61:121–129
61. Pons A, Roca P, Aguiló C, Garcia F, Alemany M, Palou A (1981) A method for the simultaneous determinations of total carbohydrate and glycerol in biological samples with the anthrone reagent. J Biochem Biophys Methods 4(3–4):227–231
62. Lowry OH, Rosebrough NJ, Farr AL, Randall RJ (1951) Protein measurement with the Folin phenol reagent. J biol Chem 193(1):265–275
63. Jagota S, Dani H (1982) A new colorimetric technique for the estimation of vitamin C using Folin phenol reagent. Anal Biochem 127(1):178–182
64. Singleton VL, Orthofer R, Lamuela-Raventós RM (1999) Analysis of total phenols and other oxidation substrates and antioxidants by means of folin-ciocalteu reagent. Methods in enzymology, vol 299. Elsevier, pp 152–178
65. Zhang Q, Ma R, Tian Y, Su B, Wang K, Yu S et al (2016) Sterilization efficiency of a novel electrochemical disinfectant against Staphylococcus aureus. Environ Sci Technol 50(6):3184–3192
66. Tian Y, Ma R, Zhang Q, Feng H, Liang Y, Zhang J et al (2015) Assessment of the physicochemical properties and biological effects of water activated by non-thermal plasma above and beneath the water surface. Plasma Processes Polym 12(5):439–449
67. Hongzhuan Z, Ying T, Xia S, Jinsong G, Zhenhua Z, Beiyu J et al (2020) Preparation of the inactivated Newcastle disease vaccine by plasma activated water and evaluation of its protection efficacy. Appl Microbiol Biotechnol 104:107–117
68. Shen J, Tian Y, Li Y, Ma R, Zhang Q, Zhang J et al (2016) Bactericidal effects against S. Aureus and physicochemical properties of plasma activated water stored at different temperatures. Sci Rep 6(1):28505
69. Rathore V, Patel S, Pandey A, Savjani J, Butani S, Dave H et al (2023) Methotrexate degradation in artificial wastewater using non-thermal pencil plasma jet. Environ Sci Pollut Res 1–10
70. Rathore V, Pandey A, Patel S, Savjani J, Butani S, Dave H et al (2024) Penicillin antibiotic (Ampicillin and Cloxacillin) degradation using non-thermal Pencil plasma jet. Water Air Soil Pollut 235(1):1–16
71. Zhao Z, Wang X, Ma T (2021) Properties of plasma-activated water with different activation time and its effects on the quality of button mushrooms (Agaricus Bisporus). Lwt 147:111633
72. Ke Z, Peng X, Jia S, Liu S, Zhou X, Ding Y (2024) Mechanisms underlying the potent antimicrobial effects of plasma-activated seawater (PASW) on fish spoilage bacterium Shewanella putrefaciens. Food Chem 140147

73. Nasiru MM, Boateng EF, Umair M, Alnadari F, Bassey AP, Qian J et al (2024) Decontamination of egg-associated pathogens by plasma-activated water and hydrogen peroxide. J Food Saf 44(3):e13136
74. Lotfy K, Khalil S (2022) Effect of plasma-activated water on microbial quality and physicochemical properties of fresh beef. Open Phys 20(1):573–586
75. Tan M, Chen W, Li M, Luo Q, Xiao Y, Liu F et al (2024) Effect of air, O2, Ar, and N2 plasma-activated water on mildewing activity of moldy pathogen of Gannan navel oranges. Plasma Processes Polym 21(3):2300179

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Authors and Affiliations

Vikas Rathore[1] · Piyush Sharma[2] · Arun Prasath Venugopal[2] · Sudhir Kumar Nema[1,3]

✉ Vikas Rathore
vikas.rathore@ipr.res.in; vikasrathore9076@gmail.com

[1] Atmospheric Plasma Division, Institute for Plasma Research (IPR), Gandhinagar, Gujarat 382428, India

[2] Department of Food Process Engineering, National Institute of Technology Rourkela, Rourkela, Odisha 769008, India

[3] Homi Bhabha National Institute, Training School Complex, Anushaktinagar, Mumbai 400094, India